

NAVONILA KARMAKAR

Final Year BS-MS Physics Student

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RESEARCH EXPERIENCE

MS Thesis Project, Max Planck Institute for Astrophysics, Germany

06/2026-Present

Supervisors: Prof. Dr. Volker Springel and Dr. Alankar Dutta

Topic: Hydrodynamic simulation under realistic cloud wind conditions, including multi-cloud interactions, to develop robust predictions for cloud survival, evolution, morphology.

Remote Project Intern under Dr. Morten Andersen, ESO, Germany

05/2025-Present

Characterizing the intermediate and low-mass stellar content of Barba 2, a newly discovered supergiant-rich stellar cluster: imaging data of H and Ks bands from Gemini F2, data reduction using Dragons-Flamongs2 and Astropy, infrared photometric (aperture) study of cluster stars using photutils, calibration with 2MASS photometric data; ongoing work for determining the masses of the individual stars and age of the cluster based on the pre-main sequence turn on mass, quantifying mass segregation in the cluster through any variations in the derived Mass Function for the cluster and more.

Astronomy Lab Course Semester Project, IISER Kolkata

04/2026-05/2026

Structural and Morphological Analysis of Galaxies using Image Statistics: Data acquisition from SDSS SkyServer for three different types of galaxies: NGC 2967 (Sc spiral), NGC 3351 (SBb barred spiral), NGC 3379 (E1 elliptical), image processing, surface brightness profile and model fitting, error and noise analysis, CAS morphological parameters determination, isophote analysis, global fourier analysis (2D FFT), radial Fourier analysis (Annular 1D FFT), correlation and statistical analysis, radial structural decomposition, final comparison and classification. ([Report](#))

LIGO GET-GO Summer Intern under Prof. Laura Nuttal & Dr. Rahul Dhurkunde

Institute of Gravitation and Cosmology, **University of Portsmouth, UK**

05/2025-08/2025

Mitigating the effect of glitches in the search of precessing systems: basics of GW searches such as matched filtering, signal consistency tests, coincident detection, ranking statistic, candidate significance etc using PyCBC; differences in waveform, template bank and ranking statistic for precessing searches; mathematical formalism of harmonic decomposition of precessing waveforms, performing precessing searches on a few days of data of O3 in CIT cluster, identification of GPS times of glitches that are affecting precessing searches, types of glitches, estimation of sensitivity for a simulated population of signals (injections) at a given FAR, comparing the sensitivity of the new model with existing general searches. ([Presentation](#))

Summer Research Fellow under Dr. Saurabh, ARIES Nainital

05/2024-07/2024

Understanding Star Formation in Young Clusters from Ground-based Observation: observation in 3.6 m Devasthal Optical Telescope, imaging and spectroscopy with TANSPEC; data reduction using IRAF, UVBRI photometric (PSF) study of NGC 2316 stars using DAOPHOT II, calibration with Landolt's standard SA 98 to determine true magnitude of stars, 2MASS near-infrared photometric data study for cluster identification, kinematic study from GAIA proper motion data for determining membership probability, astrometry using IRAF, GAIA Tool & WCSTools, isochrone fitting with intrinsic Zero-Age Main Sequence on HR Diagram of cluster stars for morphological analysis. ([Report](#))

Summer Intern under Prof Sudip Kumar Garain, DPS & CESSI, IISER Kolkata

07/2023

literature review on **Stellar Evolution: Interstellar Cloud to Black Holes and Pulsars.** ([Report](#))

EDUCATION

Indian Institute of Science Education and Research (IISER), Kolkata BS-MS Dual Degree, Major in Physics Courses:- classical mechanics, quantum mechanics, mathematical physics, thermodynamics and statistical physics, electrodynamics, computational physics, electronics theory & lab, nuclear physics lab, introductory astrophysics, general relativity, quantum field theory, condensed matter theory & lab, introductory cosmology, astronomy lab, fluid dynamics and MHD, atomic & optical physics, advanced optics theory & lab.	<i>10/2022 - Present</i> CGPA: 8.0
Bishnupur High School West Bengal Council of Higher Secondary Education(WBCHSE)	<i>08/2020 - 06/2022</i> Percentage: 93.8
Shibdas Central Girls' High School West Bengal Board of Secondary Education(WBBSE)	<i>01/2014 - 07/2020</i> Percentage: 93.8

PUBLICATIONS

The infrared view of the Galactic star cluster Barba 2: The low and intermediate mass content
M. Andersen, N. Karmakar, S. Millstone (In preparation)

OBSERVING PROPOSALS

Co-Investigator, "A goldilocks cluster: The structure and evolution a recently discovered Galactic Red Supergiant cluster". **Hubble Space Telescope General Observer Proposal, Cycle 34 (Submitted, 2026)**. PI: Dr. Morten Andersen, CoPI: Dr. Megan Reiter.

Co-Investigator, "Stellar mass content of a recently discovered cluster Barbá 2". **Gemini Observatory - F2, Program ID: GS-2025A-FT-204, March 2025**. PI: Dr. Morten Andersen.

SKILLS

Programming Languages and Softwares: Python, C/C++, Julia, MATLAB, L^AT_EX, IRAF, DS9, DAOPHOT II, GAIA Tool, OriginLab, Anaconda, Git, GNUPlot, PLUTO, ParaView, MESA, HEASoft.

Languages: English, Bengali, Hindi

SUMMER & WINTER SCHOOLS/WORKSHOPS/CONFERENCES

PLUTO/OpenGadget3 Symposium Max Planck Institute for Astronomy, Heidelberg, Germany	<i>06/2026</i>
CMF2IMF - The Origin of the Stellar Initial Mass Function European Southern Observatory, Garching, Germany	<i>06/2026</i>
Cosmology, Astrophysics & Astronomy Special Lecture Series by International Leading Experts, Kolkata	<i>03/2026</i>
Beyond LIGO-India: Deci-Hz GW Space Mission, IISER Kolkata	<i>03/2026</i>
Glorious Women Online Symposium, India (Online)	<i>03/2026</i>

LIGO GET-GO Student Conference, University of Glagow, UK (Presentation)	<i>07/2025</i>
SOKENDAI Asian Winter School, NAOJ & SOKENDAI, Japan	<i>02/2025</i>
Sagan Summer Workshop, NASA Exoplanet Science Institute, Caltech	<i>07/2024</i>
Lecture Series on Physics of Inflation & Reheating, Raman Research Institute	<i>07/2024</i>
Summer School on Gravitational-Wave Astronomy, ICTS Bangalore	<i>07/2024</i>
REYES student (virtual), University of California	<i>06/2024-07/2024</i>
Summer School in Astronomy & Astrophysics, IUCAA Pune	<i>05/2024-06/2024</i>
Workshop on Gravity, Cosmology and Raychaudhuri's Equation, IUCAA & JU	<i>03/2024</i>
18th Kavli Asian Winter School on Strings, Particles and Cosmology Yukawa Institute for Theoretical Physics and ICTP (online attendee)	<i>12/2023</i>
10th IAPT National Student Symposium on Physics, Chandigarh	<i>08/2023</i>
START (Space Science and Technology Awareness Training), ISRO	<i>07/2023-08/2023</i>

HONOURS AND AWARDS

MPG-IISER MS Project Research Fellow	<i>06/2026-Present</i>
Fellowship sponsored by Max Planck Institute for Astrophysics , selected through MPA Internship program & MPG-IISER program	
LIGO GET-GO Summer Research Fellow	<i>05/2025-08/2025</i>
Gravitational wave detector Expertise and Technology-GrOwth (GET-GO) fellowship sponsored by University of Glasgow , selected through LIGO SURF program	
IASc-INSa-NASI Summer Research Fellow	<i>05/2024-07/2024</i>
Summer Research Fellowship Program (SRFP) sponsored by Indian Academy of Sciences (Bangalore), Indian National Science Academy (New Delhi), National Academy of Sciences, India (Prayagraj)	
Silver Honour (Youth), IAAC (International Astronomy and Astrophysics Competition)	<i>06/2023</i>
Bronze Medalist (Senior), National Level Astrophysics Quiz Competition IAPT and A.P.J. Abdul Kalam National Council of Young Scientists	<i>09/2021</i>